

Multidimensional Attribution of Coral Bleaching Mortality on the Great Barrier Reef

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[Abstract Placeholder]

Introduction

Climate change is increasing the frequency and intensity of marine heatwaves, causing widespread coral bleaching and mortality across the Great Barrier Reef (GBR) (Eakin, Sweatman, and Brainard 2019). As thermal stress shifts from acute events to chronic disturbance, managers must transition from monitoring to targeted spatial interventions (Oppen et al. 2017). Deploying resilience-based strategies—like localized solar radiation management, assisted gene flow, or culling benthic competitors—requires accurate attribution of coral mortality to specific thermal thresholds (Hughes et al. 2018; Oppen et al. 2017). Treating the GBR uniformly masks the spatial variation in bleaching and survival observed among neighboring reefs (Wold et al. 2023). Directing conservation resources at actionable scales requires spatially explicit models that identify the environmental and biological factors moderating survival during severe heatwaves.

Current diagnostic tools rely on static or coarse-scale environmental proxies that miss fine-scale physical and ecological mediators of thermal stress (Safaie et al. 2018). Many vulnerability assessments evaluate Degree Heating Weeks (DHW) as an isolated, linear driver of mortality without adjusting for background coral recovery (MacNeil et al. 2019). This omission conflates baseline demographic turnover with bleaching-attributable loss, biasing mortality estimates. Most models also omit localized optical and hydrodynamic modifiers—such as water clarity (Secchi depth), cloud cover shading, and current flushing speeds—or shifts in community composition, such as the proportion of sensitive *Acropora* taxa (Cacciapaglia and Woesik 2016). Deterministic forecasts trained only on historical heatwaves often fail under novel thermal regimes. This inflexibility led to widespread underprediction of mortality during the 2024 mass bleaching event, limiting the effectiveness of adaptive management.

We developed a growth-adjusted framework to quantify coral mortality across the GBR. We combined multidecadal benthic photo-transects and manta tow data with high-resolution environmental predictors, including winter temperature variability, historical thermal exposure, and acute optical modifiers. To isolate bleaching impacts, we adjusted baseline mortality estimates using reef-specific Gompertz coral growth parameters to account for expected biological recovery between surveys. We paired machine-learning algorithms (Random Forests and Boosted Regression Trees) to capture non-linear feature interactions with Bayesian hierarchical beta regression models to bound proportional mortality. The Bayesian models propagate parameter uncertainty and partition spatial variance across reef clusters. By explicitly modeling interactions between thermal stress and optical mediators—specifically how localized Secchi depth and cloud fraction buffer DHW—we link mortality attribution to measurable physical contexts.

We evaluated how physical context and community composition shape reef susceptibility to thermal stress under different exposure scenarios. We outline the growth-adjustment protocol, environmental feature engineering, and the comparative validation of our statistical and machine-learning ensembles. Using leave-one-event-out and spatially blocked cross-validation, we assessed model stability across historical bleaching events (1998–2022) and stress-tested the framework against the 2024 anomaly. Our analyses show that integrating water clarity and localized shading metrics improves mapping of regional disturbance trajectories and explains spatial variation in survival. Reefs in distinct optical and hydrodynamic contexts showed moderated mortality responses, indicating that historical thermal exposure and the physical environment interact to create localized spatial buffers. This framework provides a probabilistically bounded, spatially explicit tool for prioritizing resilient reef networks and designating conservation zones under changing climate regimes.

Methods

Benthic Data and Growth-Adjustment

We integrated multidecadal benthic photo-transect and broad-scale manta tow data across the Great Barrier Reef spanning 1998 to 2024. This expanded dataset ($N = 465$ reef-event pairs) captures the full spectrum of recent mass bleaching events. To isolate thermal impacts from background demographic turnover, we growth-adjusted baseline mortality estimates using reef-specific Gompertz coral growth parameters. This adjustment accounts for the biological recovery expected between survey intervals. We also constructed a strictly filtered subset of confirmed bleaching mortality ($N = 305$) to evaluate model sensitivity when excluding compound disturbances such as cyclones and freshwater plumes.

Environmental Covariates and Feature Engineering

We engineered a suite of high-resolution environmental and biological predictors. Thermal stress was parameterized using Degree Heating Weeks (DHW). To test for physical buffering, we integrated localized optical and hydrodynamic modifiers, specifically acute water clarity (Secchi depth), cloud cover fraction, and prevailing current speeds. We also included winter sea surface temperature (SST) variability and the preceding-year proportion of thermally sensitive *Acropora* taxa to account for community composition.

Statistical Modeling and Validation

We applied a dual analytical architecture. We used Boosted Regression Trees (BRT) to map non-linear feature interactions and derive relative variable importance. We paired this with Bayesian hierarchical Beta and Binomial-OLRE regression models to estimate dose-response trajectories and enforce physical bounds on proportional mortality. The Bayesian framework propagates parameter uncertainty and partitions spatial variance across nested reef clusters. Model stability was assessed using leave-one-event-out and spatial sector-blocked cross-validation, and predictive skill was stress-tested on out-of-sample data from the extreme 2024 mass bleaching event.

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Results

Model Validation and Variable Importance

The dual analytical framework reliably predicted proportional coral mortality across historical bleaching eras. Spatial sector-blocked cross-validation showed the Bayesian models achieved a strong predictive R^2 and minimized mean absolute error compared to isolated univariate frameworks. Boosted Regression Trees identified Degree Heating Weeks as the primary driver of mortality. However, community composition (proportion *Acropora*) and optical modifiers (Secchi depth and cloud cover) contributed substantial relative influence, confirming that thermal stress does not act in isolation.

Dose-Response Trajectories and Environmental Buffering

Bayesian dose-response curves demonstrated that physical context strongly moderates reef susceptibility to thermal anomalies. High cloud cover and reduced water clarity (lower Secchi depth) attenuated mortality at severe DHW thresholds. This confirms that localized shading and turbidity buffer acute thermal exposure. Conversely, reefs dominated by *Acropora* in clear-water, unshaded environments experienced steeper mortality trajectories, matching the known physiological sensitivity of these taxa.

2024 Mass Bleaching Generalization

When stress-tested against the unprecedented 2024 mass bleaching event as a strict out-of-sample forecast, the framework maintained predictive accuracy. While traditional DHW-only linear models severely underpredicted 2024 mortality, integrating optical modifiers and community composition allowed the BRT and Bayesian models to map the extreme spatial heterogeneity observed during the anomaly.

Table 1: Summary of model performance metrics across validation folds.

Dataset	Model	RMSE	Predictive R2	Bias
ltmp	benchmark	0.130	0.397	-0.004
ltmp	brms	0.133	0.360	0.010
ltmp	brt	0.140	0.297	-0.017
ltmp	formal_ensemble	0.131	0.388	0.000
ltmp	stacked_ensemble	0.127	0.421	-0.001
manta	benchmark	0.159	-0.066	-0.017
manta	brms	0.141	0.169	-0.011
manta	brt	0.181	-0.376	-0.013
manta	formal_ensemble	0.141	0.169	-0.011
manta	stacked_ensemble	0.141	0.169	-0.011

Discussion

References

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